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Specification of the Reward Function for the Two-Degree-of-Freedom Robotic Arm Reinforcement Learning Environment

CP493 Directed Research Project

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1. Overview

The reward function for the two-degree-of-freedom robotic arm training environment (*ArmTaskEnv*) is a dense, shaped reward that combines five continuous penalty terms, four positive shaping bonuses, and one terminal bonus. The function is implemented in the *step()* method of the environment at *src/rl_armMotion/two_d/environments/task_env.py*, lines 457 through 480, and is evaluated at every environment step (one hundred times per simulated second at the default time step of $dt = 0.01$ s).

The reward is designed for the goal-reaching task in which the arm must (i) bring its end-effector to a target position in the workspace, (ii) align the end-effector with a target orientation, and (iii) hold that pose stably for a configurable number of consecutive steps before the episode terminates with a success bonus. The shaped form of the reward is necessary because purely terminal rewards produce a sparse signal that is very difficult to learn from on continuous-control tasks.

Functional form

The total reward at each step is the sum of three components: continuous penalties on task error and motion quality, shaping bonuses that increase as the arm approaches and occupies the goal, and a one-time terminal bonus when the hold criterion is met. Formally,

$$r_t = -\text{PENALTY_TERMS} + \text{SHAPING_BONUSES} + \text{TERMINAL_BONUS}$$

Each of the three groups is itemised below in Sections 2, 3 and 4. Section 5 defines the goal-region criteria. Section 6 explains how the goal tolerance interacts with the adaptive curriculum scheduler used during training.

2. Continuous Penalty Terms

Five penalty terms are subtracted from the reward on every step. These provide the primary gradient that pulls the policy toward goal-reaching and smooth motion.

#	Term	Coefficient	Description
P1	$-c1 * \text{goal_distance}$	$c1 = 2.0$	Euclidean distance from the end-effector to the goal position, in metres. The dominant term in the reward function.
P2	$-c2 * \text{orientation_error}$	$c2 = 1.0$	Absolute orientation error of the end-effector relative to the target orientation, in radians.
P3	$-c3 * \text{velocity_norm}$	$c3 = 0.15$	L2 norm of the joint angular velocity vector. Penalises rapid or jerky motion.
P4	$-c4 * \text{gradient_norm}$	$c4 = 0.20$	Normalised rate of change of the combined task error from one step to the next. Discourages oscillatory behaviour.
P5	$-c5 * \ \text{action}\ $	$c5 = 0.01$	L2 norm of the action vector. A small energy cost; a mild preference for low-magnitude actions.

Table 1. The five continuous penalty terms applied at every step.

Rationale

Terms P1 and P2 are the task-error terms. They are the only terms whose minimisation is necessary for success: a policy that drives them both to zero is, by definition, in the goal region. Term P3 is a smooth-motion regulariser inspired by the minimum-jerk principle of biological motor control (Flash and Hogan, 1985). Term P4 penalises high-frequency oscillation in the error signal, which is the symptom of an over-aggressive controller. Term P5 is a small action-magnitude cost; it has only marginal effect on the policy but provides a tie-breaker between otherwise equivalent actions.

3. Shaping Bonuses

Four positive bonuses are added to the reward when the corresponding condition is satisfied. These accelerate learning by providing a strong positive gradient as the arm approaches and occupies the goal.

#	Bonus	Value	When applied
B1	Progress bonus	$+1.5 * \text{progress}$	When the combined task error strictly decreased from the previous step. $\text{progress} = \text{error}_{\text{prev}} - \text{error}_{\text{curr}}$
B2	Proximity bonus	$+4.0 * (1 - d/\text{dthr})$	When the end-effector lies within 3x the goal tolerance. $\text{dthr} = 3 * \text{height_tol}$
B3	In-goal constant	+10.0	Every step the arm satisfies all three goal-region criteria (position, orientation, height)
B4	Hold growth bonus	$+2.0 * \text{hold_counter}$	Every step the arm is in the goal, growing linearly with the number of consecutive steps in goal

Table 2. The four positive shaping bonuses.

Rationale

Bonus B1 reflects classical potential-based reward shaping (Ng, Harada, and Russell, 1999): rewarding improvement leaves the optimal policy unchanged but accelerates convergence. Bonus B2 makes the reward landscape locally convex near the goal so the agent has a clear incentive to drive the last few centimetres of approach. Bonuses B3 and B4 together address the stability part of the task: once the arm enters the goal region, the constant B3 alone would let it drift in and out; the linearly-growing B4 makes it strictly better to stay than to leave and re-enter, so the agent learns to settle and hold.

4. Terminal Bonus and Episode Termination

The episode terminates as a success when the arm has held a valid goal pose for *hold_steps_required* = 20 consecutive steps, which corresponds to approximately 0.2 seconds of stable contact at the default Δt = 0.01 s. On the step that triggers termination, a single large terminal bonus is added to the reward.

Trigger	Reward effect	Episode outcome
hold_counter >= 20	+150.0 added to that step	terminated = True (success)
step_count >= 1000	no additional reward	truncated = True (timeout)

Table 3. Episode-termination conditions.

Rationale

The 150-unit terminal bonus is large enough relative to the per-step penalties (which are rarely more negative than about -5 at peak goal distance) to make goal-reaching the dominant trajectory-level signal. The 20-step hold requirement is the practical operational criterion for a successful task: a brief brush against the target without stability does not count. The 1000-step truncation limit caps episode length at 10 seconds of simulation and is in practice rarely reached by a converged policy.

5. Goal-Region Criteria

An arm pose is considered to be in the goal region if all three of the following are satisfied simultaneously. The values below are the defaults; all three are configurable at construction time and the position tolerance is dynamically adjusted by the adaptive curriculum scheduler described in Section 6.

Criterion	Default value	Configurable as
Position error < tolerance	0.10 m	goal_tolerance / height_tolerance
Orientation error < tolerance	10 degrees	orientation_tolerance_deg
Joint velocity norm < tolerance	0.30 rad/s	hold_velocity_tolerance

Table 4. The three simultaneous criteria for goal-region occupancy.

The simultaneous requirement is important: the position and orientation criteria together encode the spatial accuracy of the reach, and the velocity criterion encodes that the arm has come to rest at the target. Without the velocity criterion the agent could achieve a transient pass through the target without holding, which would not meet the physical definition of a successful goal-reaching task.

6. Interaction with the Adaptive Curriculum

The reward function itself is fixed; the practical difficulty of the task is modulated during training by the adaptive curriculum scheduler implemented in `src/rl_armMotion/two_d/training/curriculum_callback.py`. This scheduler follows the protocol of Fischer, Hoinville, Eickhoff, and Lilienthal (2021), *Scientific Reports* 11:14445, in which the position-tolerance radius is shrunk multiplicatively each time the agent's recent success rate exceeds a threshold.

Parameter	Value	Effect
initial_tolerance	0.60 m	Position tolerance at the start of training. Wide so early random exploration produces successes
min_tolerance	0.02 m	Final precision target. Tolerance is never shrunk below this value.
success_rate_threshold	80%	If the rolling success rate over the recent window exceeds this, the tolerance is decayed.
decay_factor	0.80	Multiplicative shrink applied at each decay event.
window_size	50 ep.	Number of recent episodes counted in the rolling success rate.
min_episodes_before_decay	20 ep.	Cooldown between consecutive decays, to prevent oscillation.

Table 5. Curriculum parameters and their effects on the reward function's goal criterion.

The interaction is straightforward: the curriculum calls `env.set_goal_tolerance(new_value)` at each decay event, which updates the `height_tolerance` attribute used by the goal-region check in Section 5 and by the proximity-bonus threshold in Section 3. The reward formula is unchanged. As a result, the same reward function produces a much easier task early in training (large goal radius, frequent successes, strong learning signal) and a much stricter task late in training (small goal radius, precise final convergence).

7. Historical Note on Physics Constraints

An earlier development phase of this project (commits *bf6c899* and *d89cd2b*, 21 March 2026) introduced four explicit physics-grounded reward constraints, labelled A for gravitational potential energy, C for joint acceleration, J for mechanical energy, and K for jerk, each grounded in classical mechanics and the computational-motor-control literature. These were intended to give the reward function direct biomechanical interpretability.

Across subsequent iterations the four explicit constraints were merged into the simpler, more learnable shaping form documented in Sections 2 to 4 of this report. Specifically, the jerk and joint-acceleration constraints (K and C) were absorbed into the velocity-norm penalty P3 and the gradient-norm penalty P4, the energy constraints (A and J) were absorbed into the action-norm penalty P5, and the goal-distance term (P1) was retained unchanged. The current shaped form was found to converge faster on the 2-DOF arm than the explicit-constraint form while retaining the same qualitative effect on policy behaviour. The full four-constraint form is preserved in the git history and can be restored if a future experiment requires explicit biomechanical interpretability of the reward terms.

8. Relationship to Fischer 2021's Reward Function

The reward function documented in this report was not changed as part of the integration of the Fischer et al. (2021) methodology into this project. This section explains why that was the correct choice given the current state of the environment, and identifies the reward decision as future work coupled to the muscle wiring planned in Phase 6.

8.1 Fischer's Reward Function

Fischer et al. (2021) used a deliberately simple three-term reward in their MuJoCo seven-degree-of-freedom musculoskeletal arm:

```
r_t = -|| p_end_effector - p_target || (distance penalty)
- alpha * || muscle_activation ||^2 (effort cost, alpha = 0.05)
+ R_success if distance < tolerance (sparse terminal bonus, R_success = 1.0)
```

There are no proximity bonuses, no orientation term, no gradient-norm penalty, no progress bonus, and no hold-growth bonus. The position tolerance is what their adaptive curriculum shrinks from 0.60 m to 0.02 m over training. The sparseness is intentional. Fischer's scientific claim is that human-like motion (Fitts' Law, the 2/3 Power Law) emerges from the Hill-type muscle dynamics and the adaptive curriculum, not from clever reward engineering. Adding shaping terms to the reward would weaken the emergence claim because the agent would simply be doing what the reward told it to do.

8.2 Why Our Reward Is Different

The current training environment *ArmTaskEnv* uses direct velocity-command actuation: the action vector is a normalised joint-velocity command in the closed interval minus one to plus one, applied directly to the joints with only the kinematic limits clipping the result. There is no muscle in the actuation path. Without muscle dynamics in the env, the agent has no physical regularisation that pulls toward smooth motion; the reward must supply that regularisation explicitly. Terms P3 (velocity-norm penalty) and P4 (gradient-norm penalty) in our reward exist exactly to do the smoothing work that Fischer's muscle force-velocity curve does physically. Term B2 (proximity ramp) makes the reward landscape convex near the goal so the agent can find the last few centimetres of approach without depending on a stochastic visit to the sparse success region.

If the reward were stripped to Fischer's three-term form while keeping direct-velocity actuation, two problems would arise. First, the agent would have no incentive to move smoothly: high-velocity, jerky trajectories that nevertheless reach the target would score equally well to smooth ones. The Fitts' Law and Power Law harnesses would consequently report poor numbers — but for the wrong reason. Second, the sparse success bonus alone is too weak a signal for SAC to bootstrap from in early training without the physical smoothing that muscles provide. Training would either diverge or take many more samples than is practical.

8.3 When the Reward Decision Becomes Active

Phase 6 of the Fischer integration plan, documented in *progress.md* Section 7.3, is to wire the Hill-type muscle model added in Step 5 into the training environment's step function. When that happens, the action space will become muscle activations in the closed interval zero to one rather than velocity commands. The muscle force-velocity curve will then physically regularise the motion, and the case for keeping the velocity- and gradient-norm penalties in the reward weakens. Three options will be on the table:

- **Option A.** Keep the present shaped reward unchanged. Lowest risk on convergence; weakens the emergence claim because the reward is still doing smoothing work that the muscles also do.
- **Option B.** Switch to Fischer's three-term reward exactly. Strongest scientific claim; higher risk on convergence because the 2-DOF arm has fewer muscles and lower inertia than Fischer's 7-DOF arm.
- **Option C (recommended).** Hybrid: keep distance, effort, and sparse success bonus, drop the velocity-norm, gradient-norm, and proximity-ramp shaping. Sparse enough to preserve the emergence claim; dense enough to be tractable on 2-DOF.

The choice should be made when Phase 6 lands, with actual training runs available for side-by-side comparison. The Fitts' Law and Power Law validation harnesses must in any case be re-run after the muscle wiring: their numbers will change regardless of which reward option is chosen.

8.4 Summary of the Position Taken in This Report

The present shaped reward documented in Sections 2 to 4 of this report is the principled choice *given the current direct-velocity actuation*. Fischer's three-term reward is not currently appropriate because the env lacks the physical mechanism (muscle dynamics) that supplies smoothness in Fischer's setup. The reward and the actuation layer are coupled, and the next coherent step is to change both together in Phase 6 — not to change the reward in isolation.

References

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